

Subject Description Form

Subject Code	COMP 5524
Subject Title	Workflow Management and Collaborative Systems
Credit Value	3
Level	5
Pre-requisite/Exclusion	Nil
Objectives	1. To understand the analysis, design and development of workflow and collaborative systems. 2. To identify bottlenecks and pitfalls in the systems so as to streamline intra-organizational and inter-organizational business processes. 3. Apply the Object Management Group (OMG) standards in carrying out workflow analysis, such as using Business Process Model and Notation (BPMN) standards. 4. Carry out simulation experiments on performance analysis of workflow systems using various simulation languages such as Coloured Petri Nets (CPN), and other workflow simulation software, such as AnyLogic, FlexSim and Simul8. 5. Understand the economic, social and human aspects of workflow system management.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) better understand workflow management; b) relate workflow systems to interconnect organizations to facilitate collaboration across business enterprises; c) explore office automation and workflow applications that can improve business performance; and d) perform case studies and hands-on exercises using process analysis tools.
Subject Synopsis/ Indicative Syllabus	Products and services provision in today's business; Supply chain management; IS's perspective of business process management (BPM); Maslow's hierarchy of needs; Multi-disciplinary nature of BPM; Business Process Model and Notation (BPMN 2.0) specification standard; Coloured Petri Nets(CPN); BPM lifecycle; Designation and prioritization, Process architecture, Process discovery, Process analysis, Process re-design, Process implementation and Process monitoring and controlling in BPM;
Teaching/Learning Methodology	In this course, workflow and collaborative systems will be covered from both technical and business perspectives. Case studies are used to demonstrate how to improve business performance through office automation and workflow applications. Students are expected to complete assignments in groups. This will Include case studies and hand-on software exercises. 39 hours of class activities including - lecture, tutorial, lab, workshop seminar where applicable.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	Projects	30	✓	✓	✓	✓
	Final Examination	70	✓	✓	✓	
	Total	100				
Student study effort expected	Class Contact:					
	Class activities (lecture, tutorial, lab)				39 hours	
	Other student study effort:					
	Projects, Exams				66 hours	
	Total student study effort				105 hours	
Reading list and references	1. Marlon Dumas, Marcello La Rosa, Jan Mendling and Hajo A. Reijers, Fundamentals of Business Process Management, Springer 2018. 2. Thomas Allweyer, BPMN 2.0 : Introduction to the standard for business process modeling, 2016, Books on Demand. 3. Coloured Petri Nets, https://cs.au.dk/cpnets/. 4. Manuel Languna and Johan Marklund, Business Process Modeling, Simulation and Design, CRC Press, 2018.					